Fiscal Year:	FY 2004	**Task Last Updated:**	FY 06/11/2004
PI Name:	Moore, Steven T. Ph.D.		
Project Title:	Validation of Centrifugation as a Countermeasure for Otolith Deconditioning During Spaceflight		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Physiology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	s.moore@cqu.edu.au	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	212-241-1943
Organization Name:	Mount Sinai School of Medicine		
PI Address 1:	Human Aerospace Laboratory		
PI Address 2:	Department of Neurology		
PI Web Page:			
City:	New York	**State:**	NY
Zip Code:	10029	**Congressional District:**	14
Comments:	NOTE: PI moved to Central Queensland University, Australia, July 2016.		
Project Type:	Flight	**Solicitation / Funding Source:**	99-HEDS-03
Start Date:	07/02/2001	**End Date:**	07/01/2005
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	4	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	McCollum, Suzanne	**Contact Phone:**	281 483-7307
Contact Email:	suzanne.g.mccollum@nasa.gov		
Flight Program:	Pre/Post Flight		
Flight Assignment:	STS-118 Note: Original end date changed from 7/1/2007.		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Cohen, Bernard (Mount Sinai School of Medicine) Kaufmann, Horacio (Mount Sinai School of Medicine) Diedrich, Andre (Vanderbilt University) Biaggioni, Italo (Vanderbilt University) Clement, Gilles (CNRS Toulouse France)		
Grant/Contract No.:	None		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	The Critical Design Review (CDR) is a technical review of the detailed design of the experiment to determine the compliance of the completed design with the science and mission requirements. A CDR shall be conducted when the detailed design is approximately ninety percent complete.		
Rationale for HRP Directed Research:			

Research Impact/Earth Benefits:	Results from this study are also relevant to studies of imbalance and orthostatic intolerance on Earth. Many of the postural and locomotor deficits observed in astronauts post-flight are similar to those seen in patients with vestibular disease, and findings from this study may shed light on the otolithic basis for these conditions.
Task Progress:	Pre-flight data was obtained from an astronaut subject at Mt Sinai Medical Center in December 2003.
Bibliography Type:	Description: (Last Updated: 09/07/2020)
Abstracts for Journals and Proceedings	Moore, S.T.; MacDougall, H.; Clark, J.; Wuyts, F.; Lesceu, X.; Speyer, J-J.; Cohen, B. "Spatial disorientation – how the brain interprets linear acceleration during flight." XXIII International Congress of the Barany Society, Paris, France, July 7th-9th, 2004. J Vest Res , Jul-2004
Abstracts for Journals and Proceedings	Moore, S.T.; MacDougall, H.; Wuyts, F.; Lesceu, X.; Speyer, J-J.; Cohen, B. "Head-eye coordination during simulated Orbiter landings." Aerospace Medical Association 75th Annual Scientific Meeting, Anchorage, Alaska, May 2nd-6th, 2004. Proceedings AsMA Scientific Meeting 2004. , May-2004
Abstracts for Journals and Proceedings	MacDougall, H.; Moore, S.T.; Cohen, B. "Ambulatory monitoring of high frequency otolith stimulation." 34th Annual Society for Neuroscience Meeting, San Diego, October 23rd-27th, 2004. SFN Abstracts , Oct-2004
Articles in Peer-reviewed Journals	Moore, S.T.; Clement, G.; Dai, M.; Raphan, T.; Solomon, D.; Cohen, B. "Ocular and perceptual responses to linear acceleration in microgravity: alterations in otolith function on the COSMOS and Neurolab flights." J Vest Res 13(4-6):377-393, 2003. , Dec-2003
Articles in Peer-reviewed Journals	Moore, S.T.; Diedrich, A.; Biaggioni, I.; Kaufmann, H.; Raphan, T.; Cohen, B. "Artificial gravity: a possible countermeasure for post-flight orthostatic intolerance." Acta Astronaut, in press. , Jan-2005
Articles in Peer-reviewed Journals	Moore, S.T.; Cohen, B.; Raphan, T.; Berthoz, A.; Clément, G. "Spatial orientation of optokinetic nystagmus and ocular pursuit during orbital space flight." Exp Brain Res, in press. , Jan-2005
Articles in Peer-reviewed Journals	Zhu, D.; Moore, S.T.; Raphan, T. "Robust and real-time torsional eye position calculation using a template-matching technique." Comput Methods Programs Biomed 74(3):201-209, Jun 2004. , Jun-2004